

DSA 8301: Statistical Inference for Big Data

CAT 2 Marking Scheme

Master of Science in Data Science & Analytics

20th May 2026

1. Define and explain the following modes of convergence of random variables. Provide the formal definition, notation, and a brief intuition for each case.

(a) Convergence in probability (1 marks)

Solution

A sequence of random variables $\{X_n\}$ converges in probability to a random variable X if

$$P(|X_n - X| > \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

for every $\varepsilon > 0$.

Notation:

$$X_n \xrightarrow{P} X.$$

Intuition: The probability that X_n differs from X by more than any fixed amount becomes arbitrarily small as n increases.

(b) Almost sure convergence (1 marks)

Solution

A sequence of random variables $\{X_n\}$ converges almost surely to X if

$$P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1.$$

Notation:

$$X_n \xrightarrow{a.s.} X.$$

Intuition: For almost every outcome in the sample space, the sequence X_n converges to X .

(c) Convergence in distribution (1 marks)

Solution

A sequence of random variables $\{X_n\}$ converges in distribution to X if

$$F_{X_n}(x) \rightarrow F_X(x)$$

at every continuity point of the distribution function $F_X(x)$.

Notation:

$$X_n \xrightarrow{d} X.$$

Intuition: The distribution of X_n approaches the distribution of X as n increases, even though individual realizations may not converge.

2. Suppose $X_1, X_2, \dots, X_n$ are independent and identically distributed random variables with:

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty$$

and define the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

- (a) State the Weak Law of Large Numbers (WLLN). (2 marks)

Solution

The Weak Law of Large Numbers states that if $X_1, X_2, \dots, X_n$ are i.i.d. random variables with finite mean $E[X_i] = \mu$ and finite variance σ^2 , then the sample mean converges in probability to the population mean:

$$\bar{X}_n \xrightarrow{p} \mu.$$

Equivalently,

$$P(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0 \quad \text{for every } \varepsilon > 0.$$

- (b) Using Chebyshev's inequality, prove that: (4 marks)

$$\bar{X}_n \xrightarrow{p} \mu$$

Solution

To prove that $\bar{X}_n \xrightarrow{p} \mu$ using Chebyshev's inequality, we start by calculating the variance of the sample mean $\bar{X}_n$.

Since $X_1, X_2, \dots, X_n$ are independent and identically distributed with variance σ^2 , the variance of the sample mean is given by:

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

Now, we apply Chebyshev's inequality to the random variable $\bar{X}_n$:

$$P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2/n}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}.$$

As $n \rightarrow \infty$, we have:

$$P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

Therefore, by definition of convergence in probability, we conclude that:

$$\bar{X}_n \xrightarrow{p} \mu.$$

- (c) Explain the practical significance of the WLLN in statistical estimation. **(2 marks)**

Solution

The practical significance of the Weak Law of Large Numbers in statistical estimation is that it provides a theoretical foundation for using sample means to estimate population means. It guarantees that as the sample size increases, the sample mean will get closer to the true population mean with high probability. This justifies the common practice of using the average of observed data to make inferences about the underlying population parameters, ensuring that our estimates become more accurate as we collect more data.

3. Let

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2),$$

where σ^2 is known and μ is unknown.

- (a) Write down the probability density function of the normal distribution. **(1 marks)**

Solution

The probability density function of a normal distribution with mean μ and variance σ^2 is given by:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

- (b) Derive the likelihood function for μ . **(2 marks)**

Solution

The likelihood function for μ given the observed data $X_1, X_2, \dots, X_n$ is the product of the individual probability density functions:

$$L(\mu) = \prod_{i=1}^n f(X_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i - \mu)^2}{2\sigma^2}\right).$$

This can be simplified to:

$$L(\mu) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2\right).$$

- (c) Obtain the log-likelihood function. **(1 marks)**

Solution

The log-likelihood function is obtained by taking the natural logarithm of the likelihood function:

$$\ell(\mu) = \log L(\mu) = \log \left(\left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \right) \right).$$

This simplifies to:

$$\ell(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2.$$

(d) Derive the score function.

(2 marks)

Solution

The score function is the derivative of the log-likelihood function with respect to μ :

$$U(\mu) = \frac{\partial \ell(\mu)}{\partial \mu} = -\frac{1}{2\sigma^2} \cdot 2 \sum_{i=1}^n (X_i - \mu)(-1).$$

This simplifies to:

$$U(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu).$$

(e) Obtain the maximum likelihood estimator (MLE) of μ .

(2 marks)

Solution

To find the MLE of μ , we set the score function equal to zero and solve for μ :

$$U(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu) = 0.$$

This implies:

$$\sum_{i=1}^n X_i - n\mu = 0 \quad \Rightarrow \quad n\mu = \sum_{i=1}^n X_i \quad \Rightarrow \quad \hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Therefore, the MLE of μ is the sample mean:

$$\hat{\mu} = \bar{X}_n.$$

(f) Verify that the estimator corresponds to a maximum using the second derivative test

(1 marks)

Solution

The second derivative of the log-likelihood function with respect to μ is:

$$\frac{\partial^2 \ell(\mu)}{\partial \mu^2} = -\frac{1}{\sigma^2} \sum_{i=1}^n (-1) = -\frac{n}{\sigma^2}.$$

Since $\sigma^2 > 0$, we have:

$$\frac{\partial^2 \ell(\mu)}{\partial \mu^2} = -\frac{n}{\sigma^2} < 0.$$

This indicates that the log-likelihood function is concave at $\hat{\mu}$, confirming that $\hat{\mu}$ corresponds to a maximum.

4. Two insurance companies wish to compare the average number of claims processed per week.

A random sample from Company A produced: 42, 38, 45, 40, 44, 39, 41, 43

A random sample from Company B produced: 36, 35, 39, 37, 34, 38, 36, 35

Assume the population variances are equal.

At the 5% significance level, test whether the average number of claims processed differs between the two companies.

- (a) State the hypotheses. (1 marks)

Solution

The hypotheses for the test are:

$H_0 : \mu_A = \mu_B$ (the average number of claims processed is the same for both companies)

$H_1 : \mu_A \neq \mu_B$ (the average number of claims processed differs between the two companies)

- (b) Compute the sample means and sample variances. (3 marks)

Solution

For Company A:

$$\bar{X}_A = \frac{42 + 38 + 45 + 40 + 44 + 39 + 41 + 43}{8} = 41.5,$$

$$\sum_{i=1}^8 (X_i - \bar{X}_A)^2 = (42 - 41.5)^2 + (38 - 41.5)^2 + (45 - 41.5)^2 + (40 - 41.5)^2$$

$$+ (44 - 41.5)^2 + (39 - 41.5)^2 + (41 - 41.5)^2 + (43 - 41.5)^2 = 42,$$

$$S_A^2 = \frac{1}{8-1} \sum_{i=1}^8 (X_i - \bar{X}_A)^2 = \frac{42}{7} = 6.00.$$

For Company B:

$$\bar{X}_B = \frac{36 + 35 + 39 + 37 + 34 + 38 + 36 + 35}{8} = 36.25,$$

$$\sum_{i=1}^8 (X_i - \bar{X}_B)^2 = (36 - 36.25)^2 + (35 - 36.25)^2 + (39 - 36.25)^2 + (37 - 36.25)^2$$

$$+(34 - 36.25)^2 + (38 - 36.25)^2 + (36 - 36.25)^2 + (35 - 36.25)^2 = 19.5,$$

$$S_B^2 = \frac{1}{8-1} \sum_{i=1}^8 (X_i - \bar{X}_B)^2 = \frac{19.5}{7} = 2.79.$$

- (c) Compute the pooled variance. (1 marks)

Solution

The pooled variance is calculated as follows:

$$S_p^2 = \frac{(n_A - 1)S_A^2 + (n_B - 1)S_B^2}{n_A + n_B - 2} = \frac{(8-1)(6.00) + (8-1)(2.79)}{8+8-2}.$$

$$S_p^2 = \frac{42 + 19.5}{14} = \frac{61.5}{14} = 4.39.$$

- (d) Compute the test statistic. (2 marks)

Solution

The test statistic for comparing two means under the assumption of equal population variances is

$$t = \frac{\bar{X}_A - \bar{X}_B}{S_p \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}}.$$

Substituting the sample statistics,

$$t = \frac{41.5 - 36.25}{\sqrt{4.39} \sqrt{\frac{1}{8} + \frac{1}{8}}} = \frac{5.25}{2.096 \sqrt{0.25}}.$$

$$t = \frac{5.25}{2.096(0.5)} = \frac{5.25}{1.048} = 5.01.$$

Therefore,

$$t = 5.01.$$

- (e) Determine the rejection region. (1 marks)

Solution

The degrees of freedom for the test is $n_A + n_B - 2 = 14$. At a significance level of $\alpha = 0.05$ for a two-tailed test, the critical value from the t-distribution is approximately $t_{0.025,14} = 2.145$. Therefore, the rejection region is:

$$|t| > 2.145.$$

- (f) State your conclusion. (2 marks)

Solution

Since the calculated test statistic $t = 5.01$ is greater than the critical value of 2.145, we reject the null hypothesis H_0 .

Conclusion: There is sufficient evidence at the 5% significance level to conclude that the average number of claims processed differs between Company A and Company B.